

Architecture Selection: We chose **Layered Architecture** for the **Digital Campus Library Reservation Gateway**.

Reason 1 (scale and team): The system serves one campus library with 50,000 titles and a single development team. A layered design keeps the code organised (presentation, business, data) without the operational cost of running and monitoring many independent services.

Reason 2 (queue consistency): The FIFO hold queue must never give two students the same position, and cancelling a hold must move everyone up correctly. Keeping all queue data in one database behind a single Hold Manager lets us use transactions to guarantee this, which would be much harder with data split across microservices.

Security advantage: Students and librarians only reach the system through the Web UI, and every request passes through the Authentication component before it reaches business logic. The database is reachable only from the business layer, so it is never exposed directly. Librarian-only actions, such as reordering a queue, are checked by role in one place.

Performance benefit: Components in the business layer call each other directly in one deployment, so there is no network delay between services. Together with database indexing and paginated queries, this helps the catalog search meet the 150 ms target for 50,000 titles.



